

L.link: Illustrating Controllable Surprise with L-System Based Strokes

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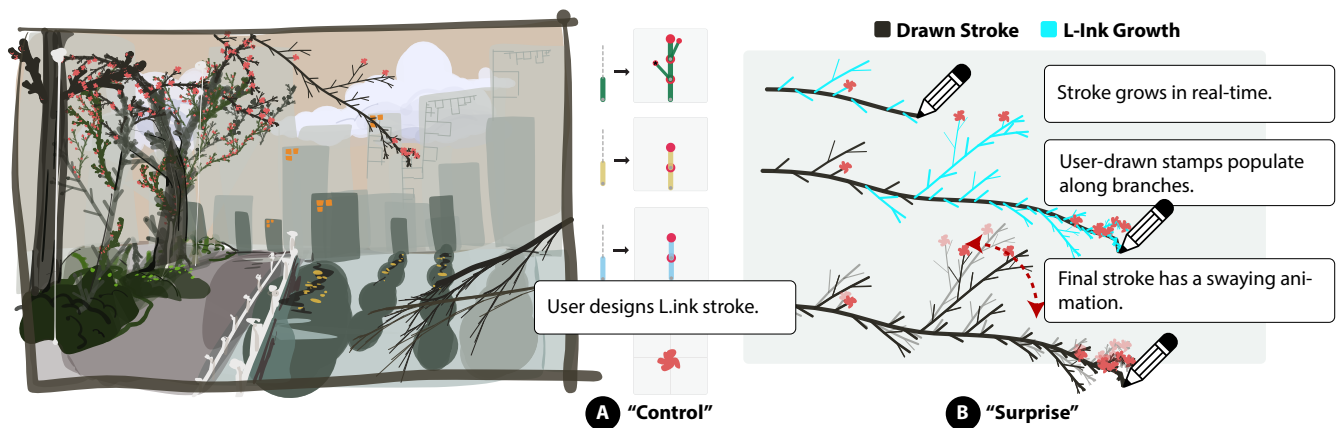


Figure 1: With L.link, users draw strokes that grow into realistic, animated, branching structures. A: Users *control* ink styles by directly manipulating branch topology, lengths, angles, and custom stamps via a visual rule editor. B: Real-time stroke growth evokes *surprise* within the user, enhanced by procedurally-arranged stamps and animations.

Abstract

Traditional usability guidelines recommend that interactive systems minimize user surprise, yet surprise plays a key role in artists' creative experience. Digital art tools thus face the challenge of balancing control and surprise. We present L.link, a digital illustration tool that empowers expert and novice artists to draw with controllable yet unpredictable procedural growth styles powered by L-systems. Unpredictability is achieved with a real-time interactive ink growth effect, while user control is provided by a rule editor with live visual preview. L.link's design builds on a model of drawing as a continuous feedback loop between the artist and their medium. Through a formative study with an early prototype version of the system, we identify three types of surprise and adapt our design with a strategy for mitigating unwanted surprise. This work has

the potential to influence future efforts towards empowering artists working with unpredictable mediums.

CCS Concepts

• **Human-centered computing** → **User interface toolkits; Web-based interaction.**

Keywords

Creativity Support Tools, Vector Illustration, Procedural Art

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1 Introduction

The principle of least astonishment asserts that interactive systems should be designed to minimize users' surprise [4]. Although this principle is desirable in many systems, it is unclear if and how it should be applied to creativity support. Indeed, surprise plays a

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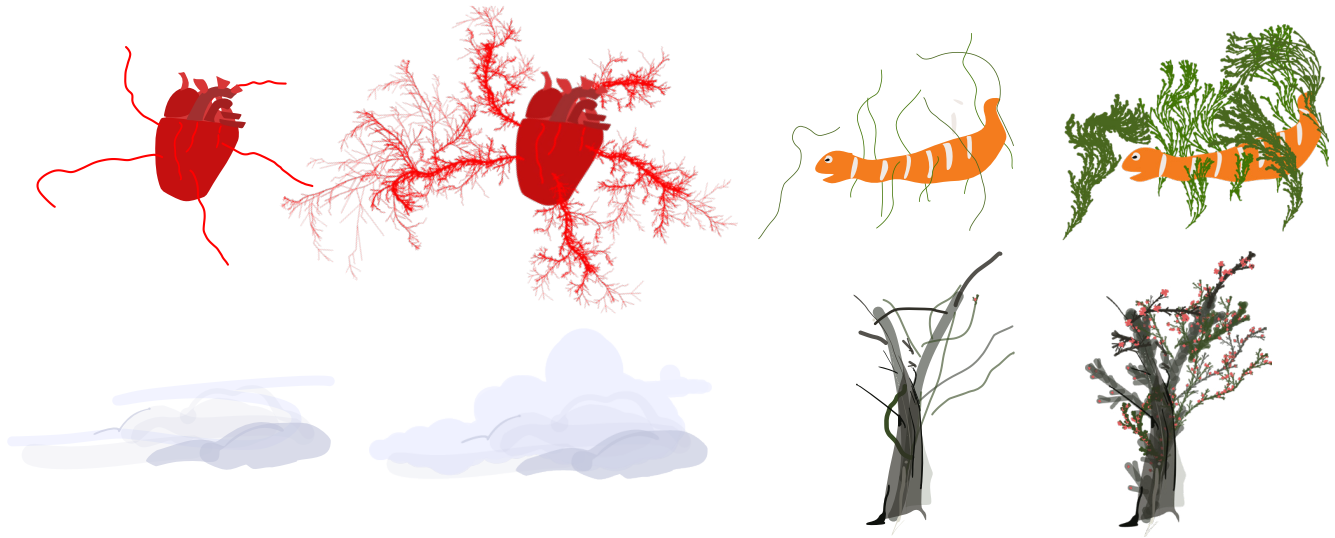


Figure 2: Comparison between sketches without L.link growth (left), and with L.link growth (right). Users can generate realistic and expressive organics with only a few strokes.

key role in the artist’s individual creation process. Cognitive psychologist Barbara Tversky describes how architectural students inspecting their own sketches react to unexpected features and relations by revising their design ideas, like “having a conversation with one’s self” [38]. The act of inspecting sketches is an example of what Schön calls “reflection-on-action,” in which discrete periods of action and reflection are interleaved. However, Schön also identifies the distinct practice of “reflection-in-action,” in which artists allow unexpected stimuli to continuously influence their work on the fly. Infinitesimal moments of surprise trigger in-the-moment thinking, which inspires further action [34]. This framework suggests that in-the-moment unpredictability may facilitate introspection, adaptation, and evolution of creative ideas.

Drawing, like other art forms, has the potential for surprise. Physical media introduce unexpected variations through the entropy of bleeding ink, interactions with paper texture, and smudging dynamics. Now with digital media, the range of possible unpredictable effects is vast—from simple physical simulation to living strokes that meander across the screen when the artist takes a break [1, 19]. By engineering unpredictable effects, these tools have the potential to transform the illustrator’s experience of reflection-in-action. Of course, simply maximizing surprise would make illustration tools unusable—some degree of control is necessary. The tension between surprise and control suggests a fruitful area of investigation.

In exploring this tension, we can observe that unpredictability and determinism are not mutually exclusive. Lindenmayer systems (L-systems), commonly used to generate realistic plant growth for graphics applications, are both deterministic and unpredictable. Given a simple set of rules, L-systems generate emergent structures which appear random and organic, yet the same rules always produce the same structures. This property implies that users could customize L-system rules with full control, yet still experience surprise upon visualizing the resulting structure.

We present L.link, a digital illustration tool empowering artists to draw with controllable yet unpredictable growing ink powered by the L-system representation. Strokes grow into intricate branching structures in real-time to enhance the natural unpredictability of L-systems and contribute to in-the-moment engagement through surprise. On the other hand, a visual rule-editor with a live preview allows users to precisely control parameters to edit and remix a customized set of ink styles. L.link’s design, informed by early formative studies, aims to explore how artists might interact with a drawing tool that is deterministic and controllable, yet highly unpredictable. Figure 2 illustrates example outputs with and without ink growth.

2 Related Work

2.1 Augmented Illustration

Digital illustration provides opportunities to augment the drawing process with computation, enabling experiences and outcomes impossible with traditional media. Computational augmentation fulfills a range of functions. CAD/CAM systems like SketchPad [37] provide enhanced precision by specifying exact constraints. Other systems enable extrapolating 2D sketches to 3D models [3, 6] or allowing sketches to be associated to external data for visualization [18, 22, 40]. Still another class of augmented illustration involves computationally simulating physical media like watercolor, oil paint, or bleeding ink [7, 9], combining realistic physical media with the convenience of digital erasure and undo operations.

Digital illustration tools also enable drawing with new, imaginative materials. For example, in Vignette [17], users draw with repeated patterns like fish scales, hair, or vegetation by duplicating fractional examples drawn by hand. Other works extend this idea by allowing strokes to represent *dynamic* materials as well. Draco [16] and Kitty [15] allow drawing kinetic textures of object collections like waving tree leaves or a swimming school of

fish. Energy Brushes [41] enables drawing with energy strokes that drive dynamic movement of user-drawn materials like fire and smoke. Filtered.ink incorporates dynamic textures directly into the user's ink, eliminating the need for post-hoc control strokes and demonstrating smoother integration into artists' drawing processes [43]. Dynamic Brushes also explores integration strategies via a block-based brush programming paradigm [14]. Finally, Neural Brushstroke Engine [35] uses data-driven deep learning methods to allow users to generate brush textures with natural language descriptions like "lady bug" or "blades of grass."

2.2 Interactive Growth Models

L-systems are procedural models that generate organic branching geometry, typically representing plants [20]. Formally, an L-system is specified by a starting symbol called the axiom and a set of textual substitution rules of the form $A \rightarrow B$, which indicates that the symbol A should be replaced with the string B at each iteration. L-system generation occurs in two phases: a *growth phase* and an *interpretation phase*. In the growth phase, we begin with the axiom symbol and repeatedly transform it by applying the substitution rules in parallel, deriving a longer string with each iteration. In the interpretation phase, the derived "L-string" is mapped to geometry by sequentially reading symbols as "turtle graphics" commands [32]. Standard L-systems as described above are fully deterministic, yet produce complex, organic-looking structures.

Since their conception, L-systems have evolved to incorporate various forms of user interaction. Early developments included real-time visual feedback while tuning L-system parameters [28–30], conditional rule application depending on live user interaction [10, 25], and post-hoc branch repositioning [31].

Of particular relevance are interfaces where 3D drawing (gestures) or 2D drawing (sketching) directly guides the progression of an L-system's growth. Onishi et al. developed a system in which 3D hand position interactively generates a trunk which can later be interactively grown into a branching structure [26]. The Sketch L-System [13] introduces a method for 3D L-system generation by 2D sketching using a special symbol whose geometric interpretation depends on the user's pen position. Drawing a stroke advances the L-system growth iteration-by-iteration with real-time visual feedback. L.ink extends the foundational ideas of this work. In the Sketch L-System, branches are unaffected by the user's motions once generated, making each branch visually identical up to differences in growth iterations. While this works well in a tool made for realistically modeling 3D plants, it limits the expressiveness and variation of 2D illustrations. L.ink makes each branch fully responsive to the user's motions, allowing significant variation between strokes utilizing the same underlying L-system.

Other interactive growth systems utilize a variety of models besides L-systems, including hierarchical structure-graphs [8], self-organizing models [21, 27], Markov random fields [5], a foliage spray painting paradigm [42], and example-based editing for novices [23, 24]. Our decision to use L-systems rather than other models was guided by their simple formulation and deterministic nature.

Our research expands upon prior technical contributions by exploring unique affordances gained by thinking of L-systems as 2D ink (opacity-varying branches, hand-drawn stamps, 2D animation,

and integration with familiar drawing controls like thickness and color), incorporating real-time visual feedback through live previews, enabling new forms of surprise through dynamic vector animations, making interactive L-systems accessible through an open web-based application, and most crucially by studying how continuous ink growth impact artists' experience.

3 Design Considerations

Our design philosophy stems from the idea that an illustrator is situated in a feedback loop with their illustration tool [36]. The design of L.ink is motivated by an attempt to balance two key properties of the feedback process: surprise and controllability. Formative user studies with artists and novices guided design choices towards achieving this goal. Two experienced artists (P2, P4) and two novices (P1, P3) tested an early prototype version of L.ink. In the prototype, ink style customization was only possible by editing textual L-system rules, and manipulating parameters like branch angle and branch thickness via sliders. Each participant joined us for a 25 minute session in which they described their drawing background, explored the prototype in a 10 minute think-aloud session, and participated in a semi-structured interview. Interview questions asked for open-ended feedback with an emphasis on the participants' *experience* while using the tool. Participants were not directly asked about controllability or surprise, as we wanted to explore whether these themes would arise organically.

3.1 Surprising the Illustrator

The first property of interest is surprise, which we define as the degree to which the user (on average) is unable to predict how a tool will respond to their actions. Without surprise, the artist's choices are not influenced by their interactions with the tool, since they receive no new information from any given stroke. The design of our interactive growth algorithm aims to magnify the natural unpredictability of L-systems by triggering branching and turning of the stroke in real-time. We hypothesized that the living, moving quality of L.ink could involve artists more deeply in the process of reflection-in-action and reckoning with their medium to discover new creative ground [34]. While using the prototype, participants reacted with various forms of surprise, which we categorize as follows.

3.1.1 Initial Surprise. Some participants reacted with joyful surprise in response to seeing our real-time growth and animation effects for the first time. However, this initial reaction likely comes from the novelty of seeing something never before seen. Therefore, it is not necessarily unique to the L.ink drawing process, and should be treated separately from other types of surprise which continue throughout the drawing process.

3.1.2 Continuous Surprise. Besides the initial moment of surprise, participants also claimed to experience ongoing surprise as they interacted with the procedural growth effect mid-stroke. P4 said about the prototype, "I think that's really fun when a tool surprises you and it's like an interaction between you and the tool... an open dialog where I do something and the tool responds in a certain way, and then that's something that you can play with." P2 claimed that they couldn't predict exactly how the branches were going to emerge from their

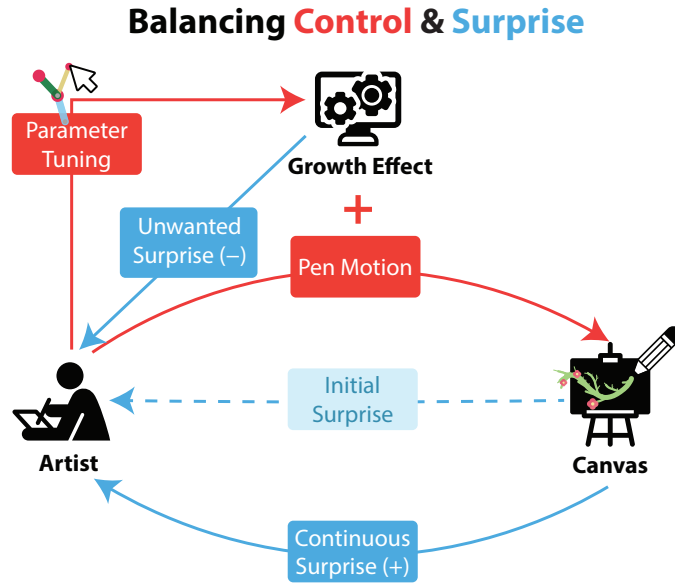


Figure 3: Blue arrows indicate surprise while red arrows indicate control. The primary feedback loop between the artist and canvas is controlled by *pen motion* and results in *continuous surprise* for the artist. *Parameter tuning* between strokes lets users alter the growth effect. We observed that *unwanted surprise* occurs when parameter tuning leads to unexpected results.

stroke at any given moment, especially when abruptly changing directions—they described this as “*exciting*” and “*organic*.” The complexity of the real-time growth algorithm and speed-sensitivity led P1 to erroneously identify it as a stochastic process, claiming that the “random” variation between strokes made the tool more “*engaging*” and “*likeable*.” These findings suggest that drawing with our L.link prototype successfully evokes a kind of continuous surprise in the user, in which each instant of growth has an element of unpredictability. While more detailed analysis is warranted to examine the impact on creative workflows, both novices and experienced artists considered this form of surprise to positively impact the drawing experience in this formative study, validating the prototype for continued development in this direction.

3.1.3 Unwanted Surprise. Not all forms of surprise were positive in this formative study—every participant reported at least one instance in which they changed an L.link parameter to try to match an image in their mind, but when they drew with the L.link, the stroke’s style was not what they wanted. This observation confirms that users desire a balance between surprise and control.

3.2 Tempering Surprise with Controllability

We define controllability as the ability to change the dynamics of the feedback loop from the outside, i.e. by tuning L.link parameters between strokes. Guided by psychological research on play, we hypothesize that effective controllability could encourage exploration by approaching the sweet spot of surprise in which the user lacks complete understanding, yet feels like they have the potential to acquire it [2]. In our formative study, participants favored the

unpredictable mid-stroke dynamics of our real-time growth algorithm, but disliked unpredictability while editing L.link parameters between strokes. Participants expressed frustration with alternating between editing and drawing in order to achieve the desired effect, and suggested adding a live visual preview to the editor to remedy this. P2 claimed that fine-tuning parameters in Blender with real-time feedback was one especially engaging part of their existing artistic workflows. P4 claimed that they like to work with interfaces that they can “*control like a tool, and fiddle and play with like a toy*.” Altogether, these responses suggest that live feedback is an essential aspect of controllability, with the potential to minimize unwanted surprise and encourage exploratory play. Figure 3 illustrates the mechanics of controllability and surprise in the feedback loop model.

4 L.link

Guided by our design considerations and incorporating feedback from the formative study, L.link introduces a continuous, interactive L-system growth technique and a visual editor enabling artists to customize ink styles with immediate feedback.

4.1 Growing Ink

4.1.1 Interactive Growth. In L.link, L-strings are pre-generated via standard (parametric) parallel rewriting [11]. Interactivity is added during geometric interpretation of the L-string symbols. As depicted in Figure 4, our method of enabling real-time growth can be thought of as “loading” the generated L-string into the pen like ink—as the user draws a stroke, symbols flow out of the pen sequentially, with some symbols depositing ink and others triggering real-time branching and turning of the stroke. In practice, this is implemented

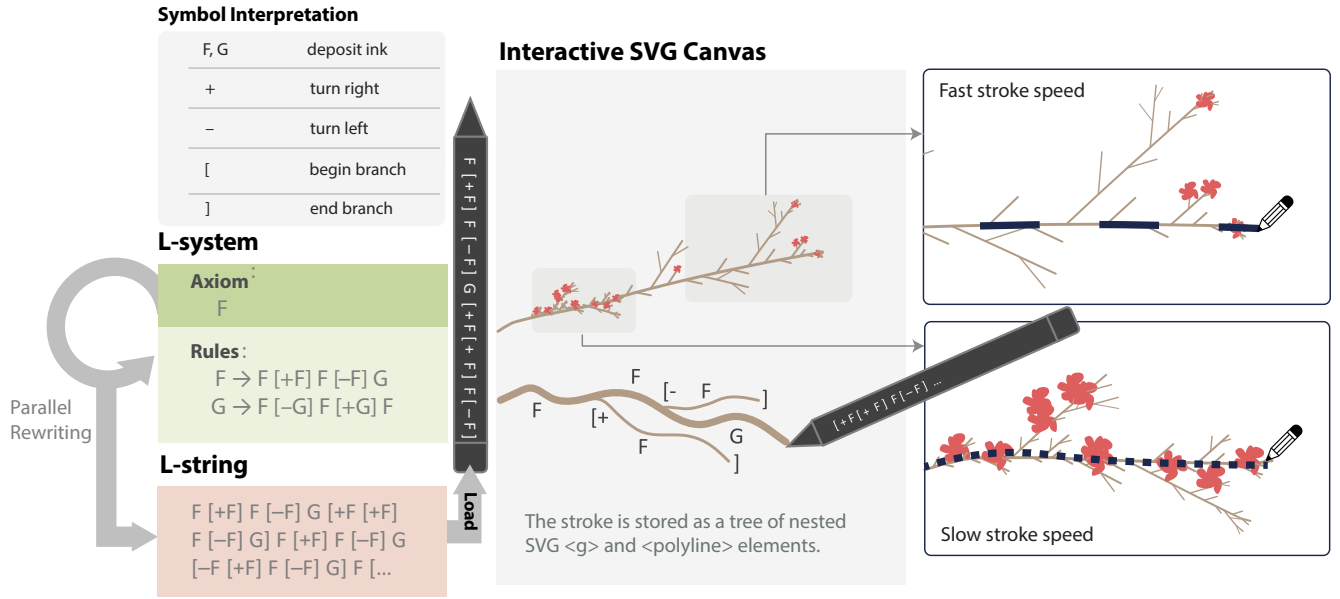


Figure 4: First, the L-string is generated by parallel rewriting. Second, the L-string is “loaded” into the pen like ink. Third, the user draws, triggering branching and turning effects as corresponding symbols are “fed out” onto the canvas. The user can vary the speed of the stroke to vary branch density.

by sampling the user’s stroke into small segments at equal time intervals and associating each segment to a single “ink symbol.” “Branch symbols” spawn new pen heads which follow the pen’s motion transformed relative to their branch’s coordinate system. This allows the user to draw curved, fluid branches, adding to stroke unpredictability. “Rotate” symbols transform the coordinate system of the branch in which they occur. Rotate symbols that occur on the base branch are ignored to prevent deviation of the stroke away from the user’s pen. With fast enough stroke sampling, growth occurs smoothly, delivering continuous surprise to the artist.

Link strokes are represented by SVG group elements. Internally, each group represents the root of a scene graph consisting of all the branches and stamps associated with that stroke. SVG transform attributes structure the scene graph while animateTransform elements enable animations. Listing 1 shows how a scene graph node (representing a single branch) is structured.

```
<g class="node" transform=...>
  <polyline points=...></polyline>
  <animateTransform additive="sum" ...></
    animateTransform>
  <g class="stamp">...</g>
  <g class="children">
    ... //child nodes
  </g>
</g>
```

Listing 1: Example SVG for a single branch of an L.ink stroke

4.1.2 Ink Animation. In order to further enhance the fluidity of the growth effect, we apply an animated scaling effect to newly drawn branches, giving them the illusion of emerging seamlessly from

their parent branch. These *ephemeral* animations last for less than a second and do not persist to the final work. Nevertheless, they enter the perception of the illustrator and thus have the potential to influence the artists’ reflection-in-action. Link also enables ongoing animations that continue after the stroke has been completed, making works created with L.ink *dynamic drawings* with *kinetic textures* [15, 16] upon SVG export. Ongoing animation uses 2 animation primitives: an animated scale (bounce) and an animated rotation (sway) which are applied to each branch individually, aided by our SVG scene-graph representation. When synchronized across a stroke, these primitives give rise to familiar swaying in the wind or pulsing kinetics. However, by de-synchronizing the motion of individual branches and linking the animation phase to the user’s pen speed, novel organic motions like *writhing* and *undulating* become possible as well.

4.2 Ink Editor

To fulfill the design goal of effective controllability, we created a rule editor, depicted in Figure 5, that exposes rules and parameters of the L-system to the user through a direct manipulation interface. We build on prior work that represents rules by color-coded segments with draggable control points [13]. However, we augment this interface in multiple ways. First, drawing inspiration from our formative study findings and Victor’s principle that “creators need an immediate connection to what they make” [39], we add a live visual preview which provides instant feedback when any parameter is changed. Second, we allow the user full freedom to modify the structure of the rules by adding and deleting branches and bends—making any parametric L-system rule reachable from any other rule via operations with real-time feedback. We hypothesize that

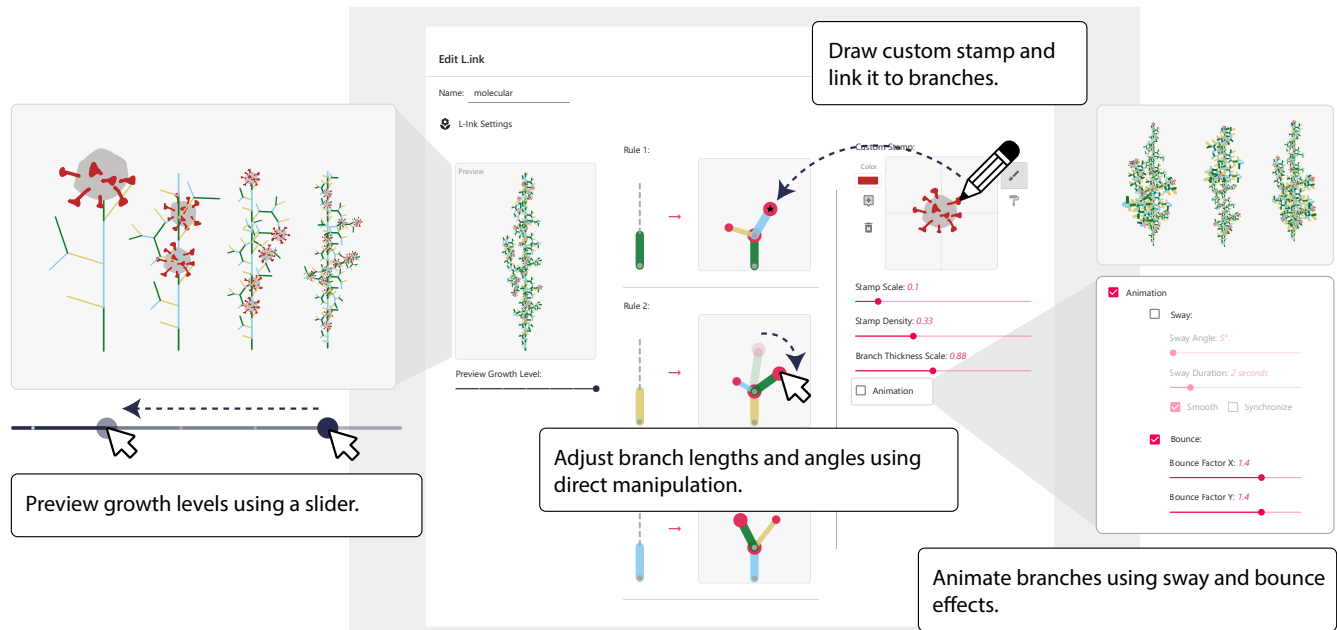


Figure 5: The L.link editor contains (inner box, from left to right): An animated live preview that reflects parameter changes immediately. A rule-editor enabling direct manipulation of control points to change rule topology, branch angles, and branch lengths. A custom stamp drawing canvas and animation menu enabling expressive bounce and sway motions.

being able to control ink parameters with immediate feedback will prevent unwanted surprise and encourage experimentation, letting the user actively explore variations of an otherwise unpredictable growth algorithm. Finally, we allow the user to add custom “stamp” symbols to the rules, enabling automatic procedural placement of hand-drawn flowers, fruits, leaves, or imaginative ornaments along the stroke. Stamps add yet another dimension of control, re-incorporating hand-drawn elements that preserve the artist’s unique style. In total, the user has control over: number of branches, placement of branch origins, number of segments per branch, angles of segments, lengths of segments, branch length scales, branch thickness scales, and custom stamps. Furthermore, the user can add sway and bounce animations with tunable magnitude, duration, smoothing, and synchronization; editing these parameters also results in instantaneous feedback to the live preview, which is itself an animated SVG element.

5 Future Work

L.link suggests promising directions for future work exploring how digital artists engage in conversation with their tools. Most importantly, a formal user study is needed to investigate how novice and expert illustrators integrate L.link into their workflows and creative experiences. Findings from such a study could provide guidelines for leveraging surprise to develop engaging tools that draw artists into the moment-by-moment experience of creativity. Furthermore, we foresee broad applicability of the themes of incomplete control and conversing with tools, especially given the increasing prevalence of generative AI. Future work could also benefit from exploring how surprise relates to other key illustration concepts or other forms of

digital art, as the strategies for achieving an appropriate balance likely depend on the medium in question.

6 Conclusion

In this paper, we present L.link, a digital illustration tool empowering artists to draw with controllably-surprising, growing ink. We demonstrate a real-time interactive growth algorithm with fluid branching, a natural stamp placement paradigm, and animations that arise seamlessly from our hierarchical SVG stroke format. Early formative studies suggest that the continuous surprise of growing ink positively impacts the creative experience. Furthermore, formative studies revealed that seamless controllability could offset unwanted surprise, guiding our creation of an L.link editor with live preview. With L.link, users can draw with expressive strokes to author and export dynamic drawings for the web. Furthermore, we believe that L.link raises important broad questions: is creativity support development harmed by the assumption that traditional usability is the optimal starting point for system design? How can artistic tools enhance artists’ embodied experience in addition to their output? Our work contributes to ongoing efforts exploring the complexity of what it means to be creative [12] and to support creativity [33]. We hope our research expands creativity support perspectives.

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